

Virtual Sensor-Guided Noise Diagnosis and Self-Correction Framework for Sensor Health Monitoring

Keyword: virtual sensing, machine learning, combustion monitoring, sensor health monitoring, noise diagnosis

Author: Siqi Dai^{1, *}

Advisor: Saurabh K Gupta^{2, *}

Undergraduate Engineering Student Competition

¹Computer, Data & Information Sciences, University of Wisconsin-Madison, WI, USA

²Mechanical Engineering, University of Wisconsin-Madison, WI, USA

^{*}Author Email: sdai66@wisc.edu, sgupta278@wisc.edu

Digital Engineering/ Virtual Sensor

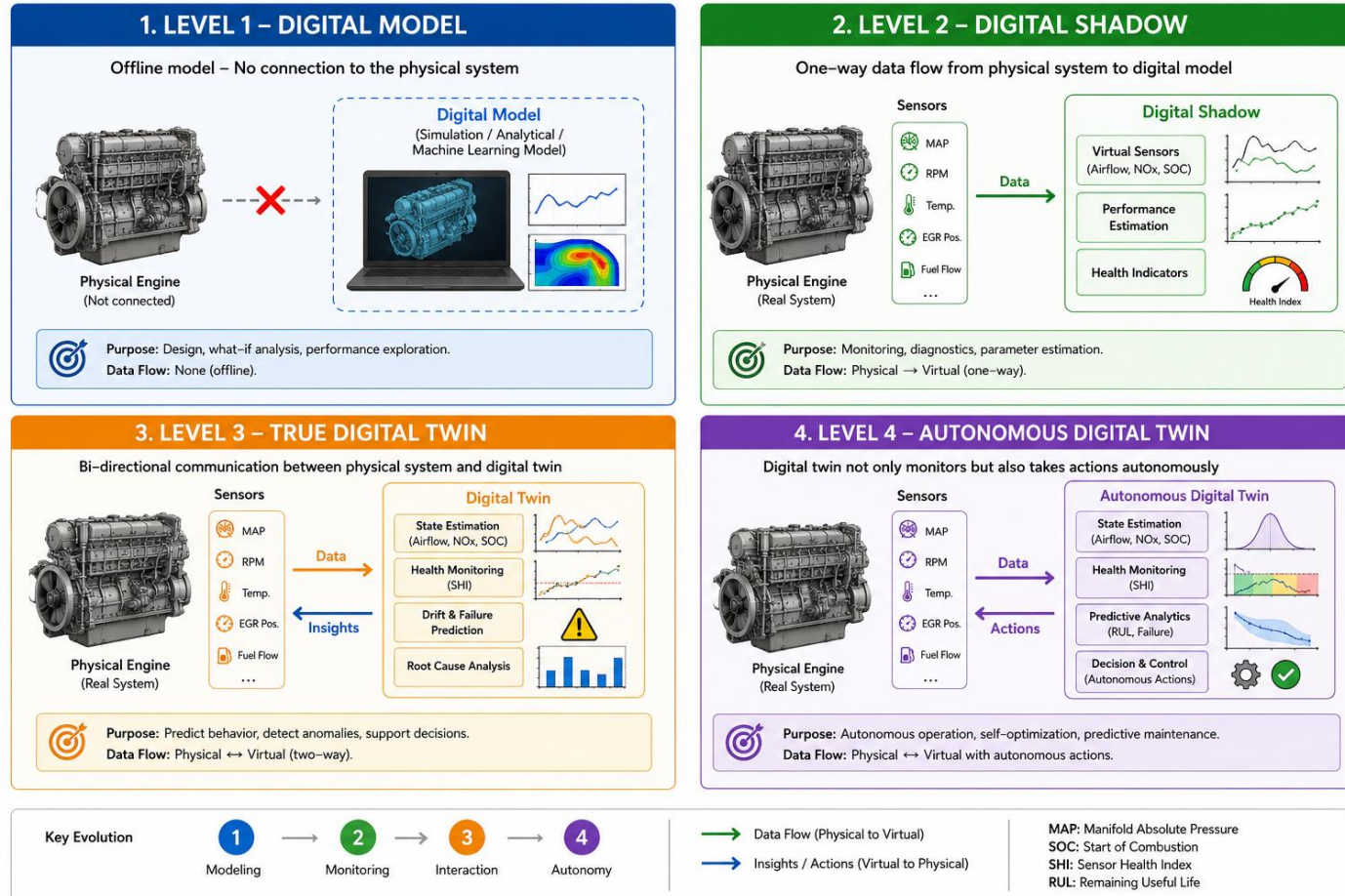


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Virtual Sensor Principle:

Model-based estimation of difficult-to-measure or important variables

➤ Physics perspective

ECU signals indirectly encode

- Air-handling dynamics
- Fuel delivery and engine load
- Thermodynamic boundary conditions

➤ Key advantage

Information recovery
without additional hardware sensors

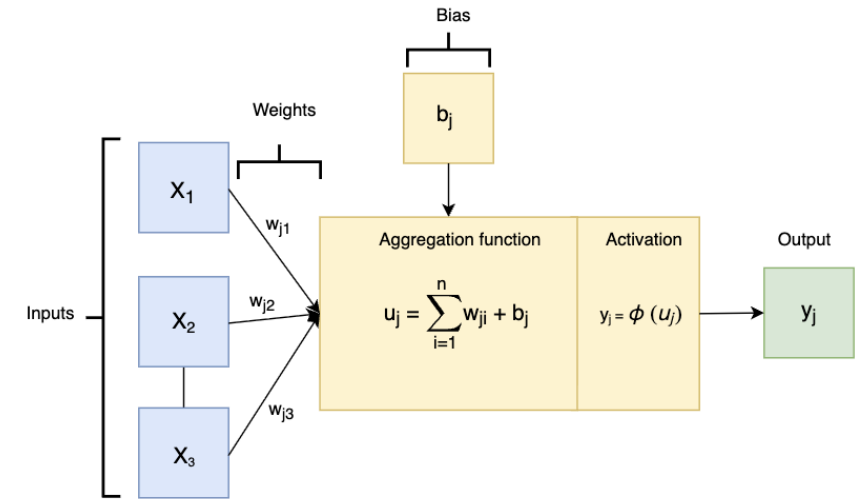
Virtual sensing shifts the challenge of hardware reliability into a robust, cost-effective software estimation solution.



Virtual Sensor Model Framework

Dataset and model

- 12-L six-cylinder heavy-duty diesel engine
- 217 steady-state operating points
- 80% training data and 20% test data
- **Six ECU inputs:** torque, ambient pressure, intake manifold temperature, intake manifold pressure, EGR rate, and VGT position
- Outputs: MF_IA, NOx_EO, and SOC
- **Structure:** $6 \xrightarrow{W^{[3]}} 64 \xrightarrow{W^{[4]}} 32 \xrightarrow{W^{[5]}} 16 \xrightarrow{W^{[6]}} 3$
- Multi-target ANN with a shared hidden representation and **ReLU activation**



Target	Metric	OLS	Random Forest	SVR (RBF)	ANN (6 inputs)
MF_IA	MAE	49.9	40.4	162.0	22.6
	R ²	0.973	0.981	0.647	0.995
NOx_EO	MAE	89.9	37.5	156.7	29.8
	R ²	0.878	0.974	0.244	0.989
SOC	MAE	1.30	0.40	0.22	0.27
	R ²	0.680	0.958	0.984	0.984

➤ The ANN achieves the highest R² for MF_IA and NOx_EO and ties the best R² for SOC. SVR gives a slightly lower SOC MAE of 0.22.



Sensitivity Analysis

Target	Input	Baseline MAE	Δ MAE (permute input)	Relative importance $I_{j,k}$
MF_IA	P_IM	22.57	181.87	1.00
	Torque		116.64	0.64
	p_0		108.14	0.59
	T_IM		83.39	0.46
	ECU_VTG_Pos		68.82	0.38
	EGR_Rate		47.36	0.26
Nox_EO	EGR_Rate	29.77	203.33	1.00
	T_IM		111.53	0.55
	Torque		58.92	0.29
	P_IM		39.86	0.20
	p_0		32.13	0.16
	ECU_VTG_Pos		26.45	0.13
SOC	Torque	0.273	1.878	1.00
	P_IM		0.763	0.41
	T_IM		0.741	0.39
	p_0		0.469	0.25
	EGR_Rate		0.458	0.24
	ECU_VTG_Pos		0.354	0.19

- Physical Consistency Established:

- P_{IM} acts as the primary driver for MF_{IA}
- EGR Rate** dominates NO_x prediction
- Torque** governs SOC

This clear hierarchy proves that a single sensor's failure will propagate uniquely through our multi-task neural network.

The neural network successfully maps thermodynamic relationships, establishing a distinct sensitivity hierarchy where specific sensor failures will dictate overall estimation health.



Fault Diagnosis Pipeline:



The complete framework closes the loop from noise-corrupted sensing to automated fault classification and self-corrected virtual-sensor output.

Fault Diagnosis Pipeline:



① Representative Case Selection

- Partition operating points into extreme percentile bands ($\leq P_{10}$, $\geq P_{90}$)
- Select the median point within each band as the representative case
- $G = \{x: x \leq Q_{10}\} \cup \{x \geq Q_{90}\}$
- $x^* = \text{median}(G)$
- Isolates stable boundary conditions; suppresses transient outliers

Representative ID	MF_IA	Torque	P_IM	EGR Rate
MF_IA_Low	221.62	840.3	24.55	11.06
MF_IA_High	1435.29	1768.9	24.55	21.44

* Sensor selection:

- P_IM is the strongest input for MF_IA prediction.
- EGR_Rate is the strongest input for NOx_EO prediction.
- Fault injection therefore focuses on P_IM and EGR_Rate at both operating points.

2 operating points $\times$ 2 sensors $\times$ 9 fault types $\times$ 19 severity levels =
684 experiments

② Sensor Fault Injection

Sensor Fault Injection Formulation

$$y_{\text{healthy}}(t) = y_0 + \varepsilon_h(t)$$
$$\varepsilon_h(t) \sim \mathcal{N}(0, [0.02|y_0|]^2)$$
$$\tilde{y}_k(t) = y_{\text{healthy}}(t) + f_k(t; \theta)$$

* The healthy signal contains 2% baseline measurement noise.

- $f_k(t; \theta)$ represents the injected fault component. The index k denotes the fault type, and θ controls the fault severity.

Virtual Sensor Residual

$$r_k(t) = \hat{y}_k(t) - \hat{y}_{\text{healthy}}(t)$$

Injected Fault Type

- Random:
Gaussian Noise, Spike
- Systematic:
Bias, Linear Drift,
Exponential Drift
- Dynamic:
Oscillation, Lag
- Hard Failure:
Dropout, Saturation

Severity Level

- Low
0.0125–0.0425
- Medium
0.0500–0.1000
- High
0.1250–0.2250
- Failure
0.2500–0.4000

5% Gaussian Noise Injection into EGR_Rate

- MF_IA_Low operating point with MF_IA as the virtual sensor target



Figure A: Healthy Sensor vs. Faulty Sensor | Input Sensor: EGR_Rate (gaussian @ Severity 0.05) -> Target Virtual Sensor: MF_IA

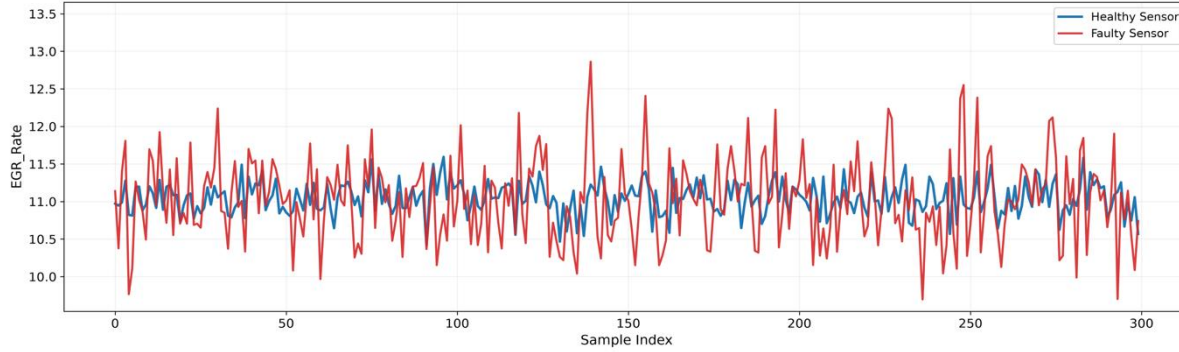


Figure B: Residual Diagnostics with Semantic Severity Bands | Input Sensor: EGR_Rate (gaussian @ Severity 0.05) -> Target Virtual Sensor: MF_IA

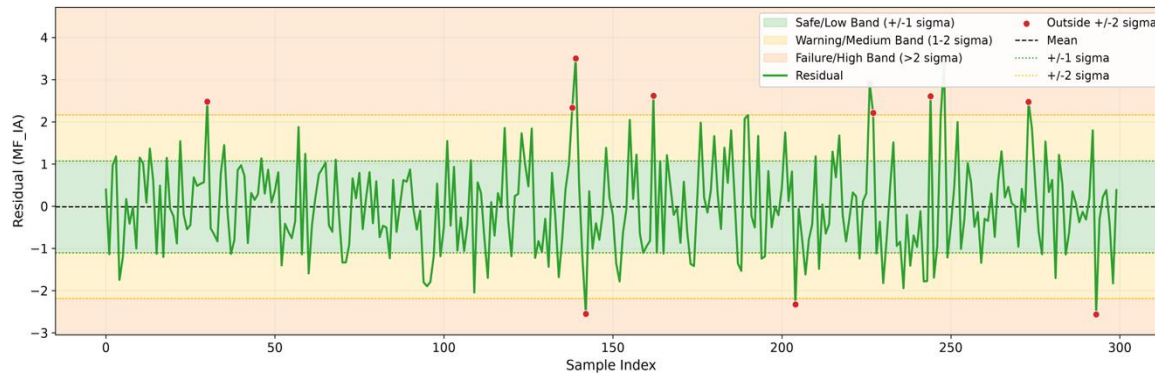
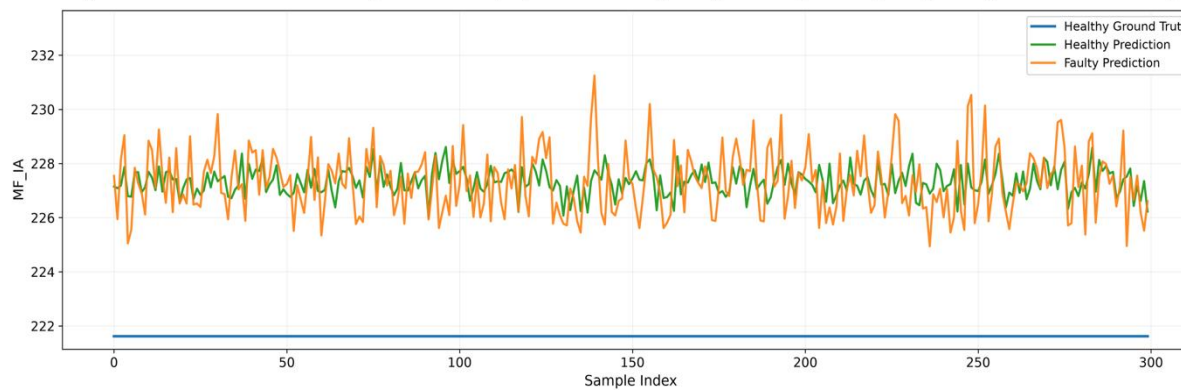


Figure C: Prediction Performance Degradation Path | Input Sensor: EGR_Rate (gaussian @ Severity 0.05) -> Target Virtual Sensor: MF_IA



Sensor Fault Injection Formulation

$$x_f(t) = x_h(t) + \varepsilon_g(t)$$

$$\varepsilon_g(t) \sim \mathcal{N}(0, [\theta | x_h(t)|]^2), \theta = 0.05$$

$$\sigma_{\text{total}} \approx |x_0| \sqrt{0.02^2 + 0.05^2} = 0.0539 |x_0|$$

Sensor Fault Injection Formulation

$$\hat{y}_h(t) = F(\mathbf{x}_h(t))$$

$$\hat{y}_f(t) = F(\mathbf{x}_f(t))$$

$$r(t) = \hat{y}_f(t) - \hat{y}_h(t)$$

$$z_r(t) = \frac{|r(t) - \mu_r|}{\sigma_r}$$

$$\begin{cases} z_r(t) \leq 1, & \text{Safe or Low} \\ 1 < z_r(t) \leq 2, & \text{Warning or Medium} \\ z_r(t) > 2, & \text{High or Failure} \end{cases}$$

Residual Response

The residual remains centered near zero, indicating little systematic prediction shift.

- Residual mean: 0.02 MF_IA units
- Residual standard deviation: 1.19 MF_IA units
- Residual range: -2.47 to 3.62
- Samples outside $\pm 2\sigma$: 13 of 300

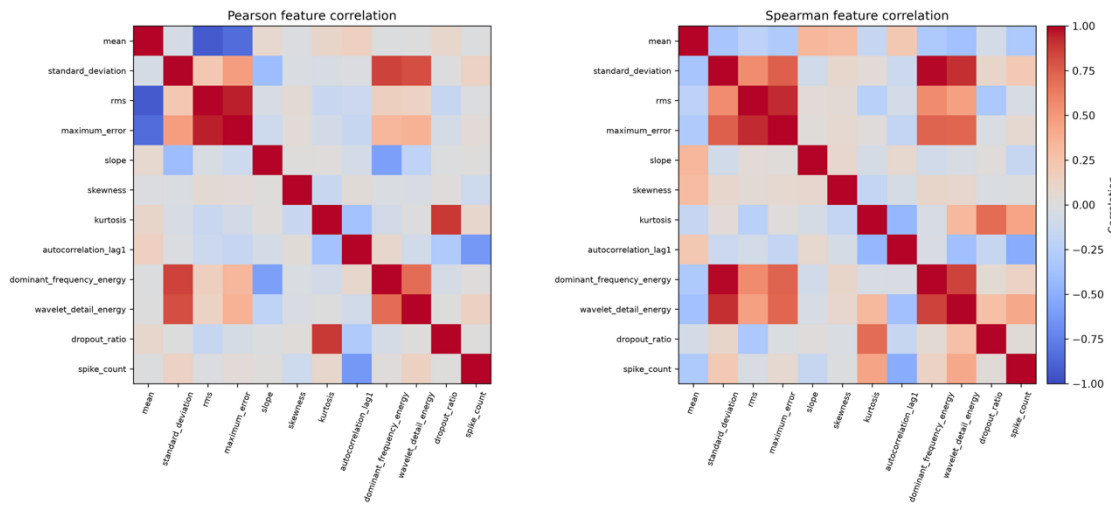


③ Leakage-Safe Classification

- Classifier input: Only the observed faulty-sensor trajectory is used. Healthy references, residuals, severity values, and fault labels are excluded from the feature set. 29 observable features

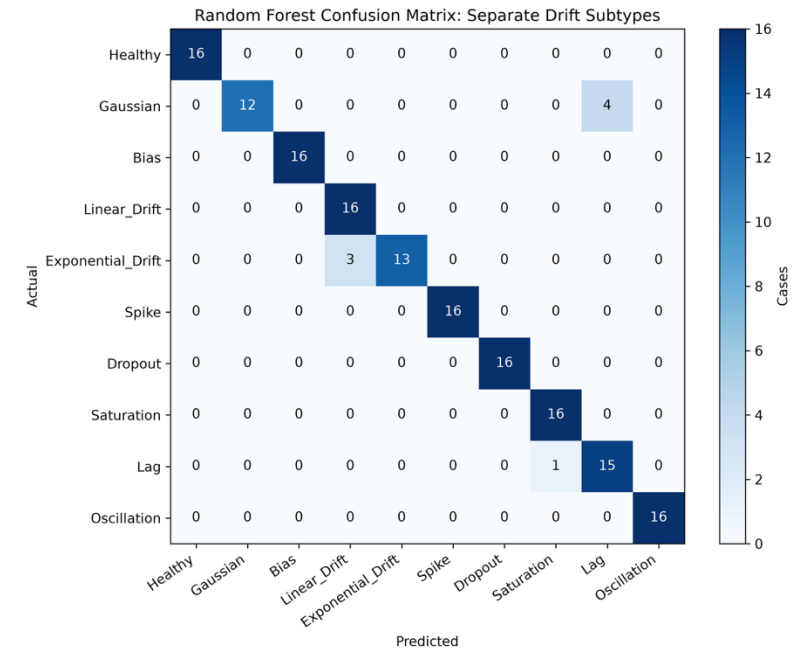
29 Observable Features

Feature Group	Examples
Distribution	Mean, standard deviation, RMS, range, quantiles
Dynamics	Slope, differences, autocorrelation, rolling statistics
Frequency	Dominant frequency, spectral energy, high-frequency energy ratio
Event Signatures	Zero fraction, robust spike count



Model & Validation

Item	Configuration
Classifier	Random forest
Validation	GroupShuffleSplit by experimental case
Training & Test	641 training, 159 test



- Using only observable sensor features with case-grouped validation, the **Random Forest** achieved **94.97% test accuracy**, demonstrating robust and leakage-safe fault classification.

Fault Diagnosis Pipeline:



④ Fault Severity Estimation

- Method
 - Residual features extracted using a 30-sample window
 - Window step of 10 samples
 - Random Forest regression
 - Fault type is not used as an input

Fault type	R ²	MAE	RMSE
Oscillation	0.797	0.027	0.036
Saturation	0.771	0.042	0.056
Exp Drift	0.763	0.041	0.053
Linear Drift	0.750	0.048	0.060
Bias	0.732	0.048	0.057
Gaussian	0.520	0.056	0.081
Spike	0.228	0.070	0.081
Dropout	0.134	0.090	0.098
Lag	-0.117	0.090	0.116

- Severity estimation works best when fault magnitude produces a smooth and continuous signal change.
- Lag severity is discrete and poorly represented by aggregate residual features.

⑤ Fault-Specific Signal Correction

Correction uses the observed faulty signal and a virtual reference estimated from the healthy signal before fault onset at sample 150.

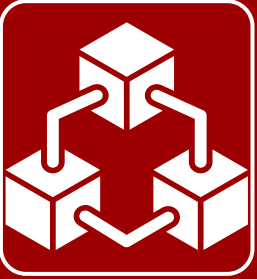
Self-correction Performance Table

Fault	Correction Method	RMSE Improvement
Bias	Median ratio correction	96.3%
Linear Drift	Polynomial ratio fitting	92.3%
Exponential Drift	Polynomial ratio fitting	93.3%
Spike	Local median and MAD with interpolation	57.6%
Oscillation	Dominant-frequency sinusoid removal	29.9%
Lag	Delay search and time shift	-8.1%

Lag correction becomes unstable at low severity because the estimated delay is only a few samples and the signal tail must be filled using the virtual reference.

Conclusion and Acknowledgments

What Works



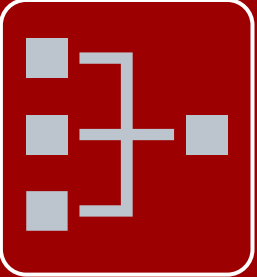
- Observable-only classification reaches 85.5% accuracy.
- Bias and drift correction reduces RMSE by more than 92%.
- Severity estimation reaches an overall R^2 of 0.487.

Consistent limitation



- Lag is the weakest fault across all three stages:
- Classification F1 0.545
- Severity estimation R^2 -0.117
- Correction improvement -8.1%

Next steps



- Introduce sequence-aware delay features and lag-specific estimation.
- Add an uncertainty threshold for ambiguous low-severity cases.
- Validate the complete pipeline under transient and unseen operating conditions.



Thank You
Questions and Discussion?

Contact: sdai66@wisc.edu